

Aditya Pun

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PROFILE

Gameplay & Tools Programmer with Distinction in a Level 7 Game Programmer Apprenticeship. Skilled in low-level systems, graphics, and performance-critical software development. Experienced in delivering commercial projects within Agile teams at AAA game studios in Unreal and custom in-house engines. Portfolio website: adipun.github.io

EDUCATION

C++ Game Programmer Level 7 Apprenticeship - Distinction

Sumo Digital | 2025

Python Data Science Course

HyperionDev | 2023

Chemistry BSc (Hons)

University of Leicester | 2015 - 2019

EXPERIENCE

Junior Programmer

Sumo Digital | **October 2023 - July 2025**

- Developed multiple real-time graphical programmes in C++ and C# using Unreal Engine and custom in-house game engines, optimising CPU, GPU and memory performance using many profiling tools.
- Created a Perlin noise terrain generator in Unreal, integrating this with Unreal's Landscapes to generate terrain meshes.
- Implemented trophies, title icon, and launch image on PS4 in line with Sony's TTR compliance, including correct metadata, system integration, and visual asset requirements.
- Developed tools in Python + Tkinter, allowing teletext visualisation and creation and management of JSON-based articles for professional writers, streamlining their workflow and introducing data validation ready for std::format.
- Collaborated in multidisciplinary teams (designers, artists, engineers) delivering milestone targets under Agile workflows (SCRUM, rapid prototyping).
- Wrote custom shaders in HLSL, created a CRT TV material shader (chromatic aberration, vignette, scanlines, pixelation), enhancing presentation, authenticity and player immersion.
- Designed a data-driven building levelling and purchasing system with CSV/JSON saving/loading in mind, improving modularity, and enabling unit testing with dependency injection.
- Optimised asset pipeline, allowing artists to add assets directly to JSON (meshes, materials, transforms), removing hardcoding and programmer dependency.

OTHER EXPERIENCE

PCR Laboratory Scientist & Technical Trainer

Rosalind Franklin Laboratory / UKHSA | **July 2021 - March 2023**

- Operated automated liquid handling robotics in a sterile, regulated environment (Containment Level 3 laboratory, ISO 35001).
- Processed 1,000+ COVID-19 tests daily, maintaining accuracy and adherence to strict quality standards.
- Trained 20+ team members on virus inactivation, sample handling (automated/manual pipetting), and quality control protocols; everyone passed compliance tests.

TECHNICAL SKILLS

- **Engines & Languages:** C++, C#, Python, Lua, HLSL, HTML/CSS; Unreal, Unity, Godot, custom in-house engines;
- **Graphics & Rendering:** Shader programming, 2D/3D mathematics, lighting, forward/deferred rendering, ray tracing, 3D modelling & transformations
- **Profiling & Optimisation:** VS Profiler, RenderDoc, VTune, PIX
- **Build, Automation & Data:** CMake, Premake, vcpkg, scripting (Python, Bash); JSON, CSV, XML, SQLite
- **Platforms & Tools:** PC (Windows), PS4 Orbis SDK, Steam Deck; Visual Studio, VS Code, Blender, Audacity, Microsoft Office Suite
- **Version Control & Practices:** Git, GitHub, Perforce; Agile (SCRUM, Waterfall), TDD, dependency injection, rapid prototyping, CI/CD, code reviews, coding standards

SOFT SKILLS

- Reliable, self-motivated, and takes ownership of tasks and challenges.
- Excellent communicator, collaborating effectively with multidisciplinary teams (artists, designers, engineers).
- Adaptable problem-solver, quick to learn new systems and work under pressure.
- Committed to continuous learning, mentoring, and knowledge sharing.
- Professional and ethical, giving/receiving constructive feedback, respecting data security, and consistently delivering high-quality work.

INTERESTS

- Making games
- Furniture restoration and DIY
- Hobbyist robotics
- Board games and D&D
- Punch needle embroidery
- Boulderling and Hiking

CREDITS

Programmer

- NUTMEG! A Nostalgic Deckbuilding
- Football Manager
- Snake Party

QA tester

- Still Wakes the Deep
- Oil Strike '75